

Ruixiang Liu

Department of Geography

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Education

- Ph.D. student, 2025.08 – present, Geography, The Pennsylvania State University, PA, USA.
- M.S., 2022 – 2025, Cartography and Geographic Information System, Wuhan University, Hubei, China.
- B.Eng., 2018 – 2022, Remote Sensing Science and Technology, Wuhan University, Hubei, China.

Research Interests

Geographic Information Science

Big Data Analytics

High-performance Geocomputing

Geospatial Data Cube

Human Mobility

Geospatial Artificial Intelligence

Peer-reviewed Journal Articles (* denotes the corresponding author)

1. Yue, P., Wang, K., Wu, H., & **Liu, R.** (2025). Recent progress and technical practices of GIServices. *Chinese Science Bulletin*.
2. Gong, J., Yue, P.*, Xiang, L., Wu, H., Wang, K., **Liu, R.**, & Teng, B. (2025). The design and implementation of the open geospatial engine (OGE). *Acta Geodaetica et Cartographica Sinica*, 54 (4), 587-602.
3. Yue, P.*, Liu, X., **Liu, R.**, Cai, C., Huang, L., Li, J., Zhang, Q., & Jiang, M. (2024). Modelling and development approaches for spatial-temporal data governance in intelligent connected vehicles. *Geomatics and Information Science of Wuhan University*.
4. **Liu, R.**, Yue, P.*, Shangguan, B., Gong, J., Xiang, L., & Lu, B. (2024). RTGDC: a real-time ingestion and processing approach in geospatial data cube for digital twin of Earth. *International Journal of Digital Earth*, 17(1), 2365386.
5. He, L.*, & **Liu, R.** (2024). Discovering links between geospatial data sources in the Web of data: the open geospatial engine approach. *ISPRS International Journal of Geo-Information*, 13(5), 143.
6. Yue, P.*, **Liu, R.**, Shangguan, B., Cao, Z., Liu, S., & Xu, H. (2023). GeoAI training data: model, quality, and services. *Geomatics and Information Science of Wuhan University*, 48(10), 1616-1631.
7. Wang, Y., Duan, Y.*, Pang, B., & **Liu, R.** (2022). The key technologies of geological disaster dynamic monitoring based on nap-of-the-object photogrammetry. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, 10, 171-176.

Conference Presentations (* denotes the corresponding author)

1. Mi, Y., **Liu, R.***, & Yue, P. A data organization approach in geospatial data cube based on variable spatial grids. In 2025 13th International Conference on Agro-Geoinformatics, Boulder, CO, USA, July 7-10, 2025. **(Oral)**
2. **Liu, R.**, Yue, P.*, Yu, D., Wang, K., Deng, H., & Li, H. A cube-enabled cloud geoprocessing engine for big Earth data. In 2023 11th International Conference on Agro-Geoinformatics, Wuhan, Hubei, China, July 25-28, 2023. **(Oral)**
3. **Liu, R.**, Yu, D.*, & Yue, P. A disaster information service for damaged road networks using dynamic segmentation. In 2022 10th International Conference on Agro-Geoinformatics, Quebec City, QC, Canada, July

11-14, 2022. (Oral, best student paper award)

International Standards

- Yue, P., **Liu, R.**, & Shangguan, B. (2024). OGC Training Data Markup Language for Artificial Intelligence (TrainingDML-AI) Part 2: JSON Encoding Standard, OGC 24-006r1. Wayland, MA: Open Geospatial Consortium Inc.
- Yue, P., **Liu, R.**, & Wang, K. (2024). OGC Training Data Markup Language for Artificial Intelligence (TrainingDML-AI) Part 3: XML Encoding Standard, OGC 24-007r1. Wayland, MA: Open Geospatial Consortium Inc.
- Yue, P., & Shangguan, B. (2023). OGC Training Data Markup Language for Artificial Intelligence (TrainingDML-AI) Part 1: Conceptual Model Standard, OGC 23-008r3. Wayland, MA: Open Geospatial Consortium Inc. (Submitter: Yue, P.[✉], Gong, J., **Liu, R.** et al.)

Teaching

As a teaching assistant:

1. GEOG 1N: Global Parks and Sustainability, The Pennsylvania State University, PA (Fall 2025)

Honors and Awards

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| 2024 | National Scholarship (top 0.2%) , China. |
| 2024 | Outstanding Student Scholarship (grade 1, top 10%) , Wuhan University, China. |
| 2023 | Outstanding Student Scholarship (grade 2, top 20%) , Wuhan University, China. |
| 2022 | Second Prize Award of Best Student Paper Competition , In the 10 th International Conference on Agro-Geoinformatics, Quebec City, Canada. |
| 2022 | Zhizhuo Scholarship (top 10%) , Wuhan University, China. |
| 2022 | Outstanding Freshman Scholarship (grade 2, top 10%) , Wuhan University, China. |
| 2022 | University Outstanding Undergraduate Thesis Award (top 5%) , Wuhan University, China. |
| 2022 | Top Ten Remote Sensing Constellations Award of Wuhan University , School of Remote Sensing Information Engineering, Wuhan University, China. |
| 2022 | First Prize , Competition on Discovering Historical Cultural Heritage with Eye in Space, Asia-Pacific Space Cooperation Organization (APSCO). |
| 2021 | Second Prize (top 15%) , MathorCup University Mathematical Modeling Challenge, China. |
| 2021 | Honorable Mention (top 15.35%) , The Interdisciplinary Contest in Modeling (ICM), The Consortium for Mathematics and its Applications (COMAP), U.S.A. |
| 2020 | Finalist (top 0.17%) , The Mathematical Contest in Modeling (MCM), The Consortium for Mathematics and its Applications (COMAP), U.S.A. |
| 2019 | Second Prize , The Chinese Mathematics Competitions (CMC), Chinese Mathematical Society, China. |
| 2019 | Excellent Student Cadre , School of Remote Sensing Information Engineering, Wuhan University, China. |

Professional Skills

- **Programming Languages:** Proficient in Scala, Java, Python; Good at C++, MATLAB; Familiar with R.
- **Programming Frameworks:** Good at Vue3 for Web design, Spring Boot for Web service, Spark for high-performance computing, PostgreSQL for data storage.
- **Professional Software:** Good at Google Earth Engine (GEE), ArcGIS, QGIS.
- **Language Skills:** Fluent in Chinese (native) and English (TOEFL score: 103 <R: 30, L: 24, S: 22, W: 27>)